a
PFOA, I = 10 mM
b
PFOA, I = 50 mM
surface-activity-controlled
electrostatic-inhibition
steric-exclusion
c
PFHxA, I = 100 mM
d
PFHxA, I = 200 mM
e
PFOS, I = 1 mM
f
PFOS, I = 10 mM
Surface excess (μmol m^{-2})
Γ_{SL}/Γ_{MA}
Concentration (mg L^{-1})
Mass action model
Langmuir isotherm
Γ_{SL}/Γ_{MA}
Experimental data
g
PFOA
PFHxA
PFOS
$\Gamma_{SL}/\Gamma_{MA} = \exp(-\Delta\Lambda_{ele})$
R^2 = 0.99954
RMSE = 0.0047
I (mM)
1
10
50
100
200
$\Delta\Lambda_{ele}$
h
PFOA
PFHxA
PFOS
Pooled empirical fit
$y_{min} = 10^{-1.79} I^{0.66}$
R^2 = 0.94
$RMSE_{log_{10}}$ = 0.12
$(\Gamma_{SL}/\Gamma_{MA})_{min}$
Ionic strength (mM)